

4. Enhancing customer experience with predictive analytics and data mining by taking suitable business scenario (Microsoft power BI).

Dataset:

- sales.csv
- products.csv
- customers.csv
- categories.csv
- employees.csv
- cities.csv
- countries.csv

Phase 1: Data Ingestion & Cleaning

► Step 1: Load Data

1. Open Power BI Desktop.
2. Click on **Home** → **Get Data** → **Text/CSV**.
3. Import all 7 CSV files one by one.
4. In the “Navigator” screen, click **Transform Data** to open Power Query Editor.

► Step 2: Rename Tables

- Rename each table in the Query Editor to:
 - Sales, Products, Customers, Categories, Employees, Cities, Countries

► Step 3: Clean the Data

For each table:

- **Remove duplicate rows:**
 - Home tab → Remove Rows → Remove Duplicates
- **Fix column names:**
 - Right-click headers → Rename (replace spaces with underscores)
- **Handle Nulls in numeric columns:**
 - In Sales: Replace nulls in quantity and unit_price with 0
 - → Select column → Transform → Replace Values
- **Convert data types:**
 - Sales[sales_date] → Data Type: Date

► Step 4: Add Date Fields

- In Sales, extract:
 - Month, Quarter, Year → Add Column → Date → Month/Quarter/Year
- Click **Close & Apply** to load cleaned data to the model.

Phase 2: Data Modeling

► Step 1: Create Relationships in Model View

Table A Column Table B Column

Table A	Column	Table B	Column
Sales	customer_id	Customers	customer_id
Sales	product_id	Products	product_id
Products	category_id	Categories	category_id
Sales	employee_id	Employees	employee_id
Customers	city_id	Cities	city_id
Cities	country_id	Countries	country_id

- Ensure relationship direction is **Single** and cardinality is **Many-to-One**.

Phase 3: Customer Segmentation

► Step 1: Create Customer Total Spend

1. Go to Sales table in Data View.
2. Create a **new column**:
 - Multiply quantity by unit_price to get the transaction amount.
 - Sum it per customer using SUMX or a calculated column.

► Step 2: Segment Customers

- Create a **new column** in Customers:
 - If total spend > 10,000 → “High Value”
 - Between 5,000 and 10,000 → “Medium Value”
 - Below 5,000 → “Low Value”

► Step 3: Build Visuals

- **Donut Chart**: % of customers in each segment
→ Fields: Customer Segment, Customer Count
- **Map**: Customers by country
 - Use Customer → Join with Cities → Countries
 - Plot country with count of customers
- **Multi-row card**: Show Top 5 customers by total spend
→ Use table visual with sort and Top N filter

Phase 4: Product & Category Insights

► Step 1: Matrix

- Rows = Category Name
- Columns = Product Name
- Values = Sum of Revenue (Quantity * Unit Price)

► Step 2: Bar Chart

- Top 10 Products by total quantity sold
→ Sort by Sum(Quantity) descending

► Step 3: Slicer

- Add a slicer for Category Name to control all visuals

► Step 4: Conditional Formatting

- In product table/bar chart, highlight products where total quantity sold < 100 using rules

Phase 5: Employee Sales Analytics

► Step 1: Table Visual

- Show:
 - Employee Name
 - Count of Orders (Order ID)
 - Total Revenue (Sum of Quantity * Price)
 - Unique Customers Served

► Step 2: Bar Chart / Treemap

- Compare employee sales visually

► Step 3: KPI Cards

- Highest revenue employee (use MAX + LOOKUP)
- Highest customer reach (use COUNT DISTINCT + MAX)

Phase 6: Forecasting Sales

► Step 1: Line Chart

- Axis = Month
- Values = Total Sales Amount (sum of Quantity * Unit Price)
- Enable Forecast in the **Analytics** pane:
 - Forecast: 3 Months
 - Confidence Interval: 95%

► Step 2: Smart Narrative

- Insert Smart Narrative to auto-explain trend and forecast

Phase 7: Business KPIs

► Create the following measures:

1. **Total Revenue:** Sum of quantity * price
2. **Average Order Value:** Total revenue ÷ total orders
3. **Customer Count:** Count of distinct customers
4. **Returning Customer %:** Customers who bought more than once ÷ total customers
5. **Sales by Region:** Group sales by Country

► Visuals:

- KPI cards for each measure
- Donut chart for regional revenue share

Phase 8: Final Dashboard & Storytelling

► Create 4 Dashboard Pages:

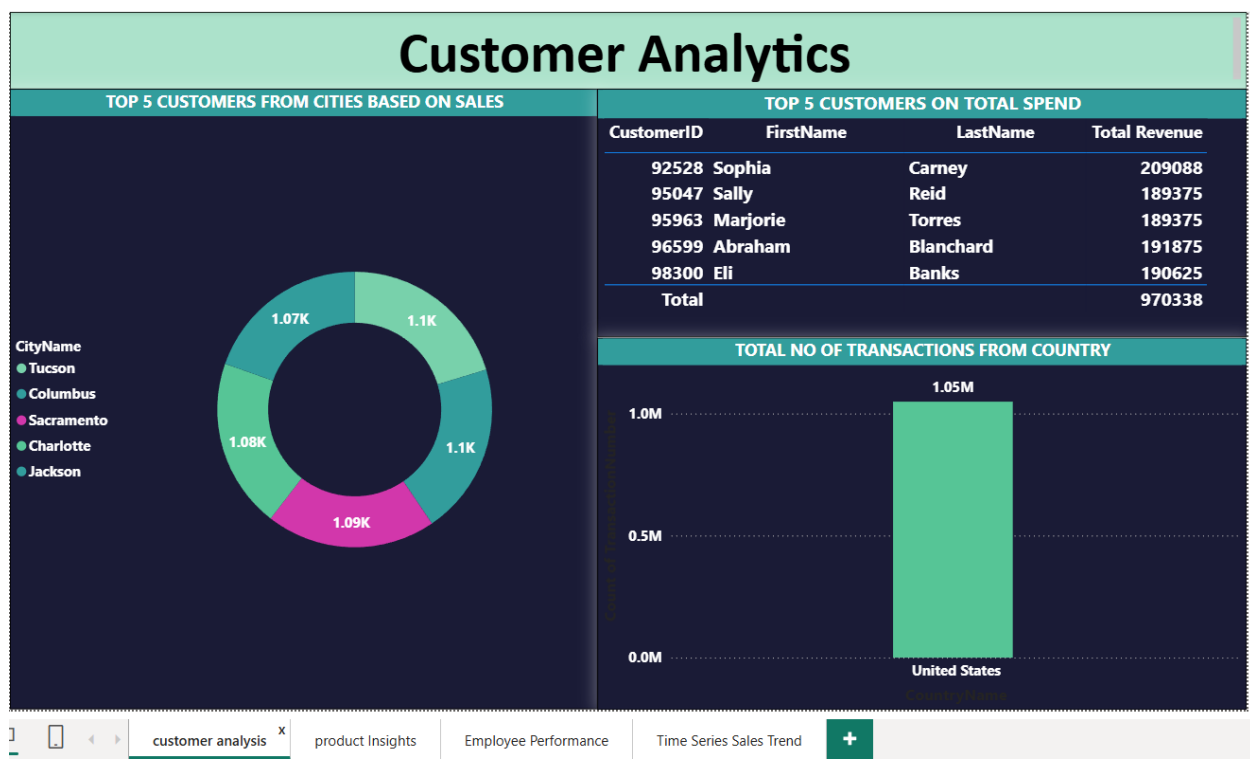
1. Customer Analytics
2. Product Insights
3. Employee Performance
4. Forecast & KPIs

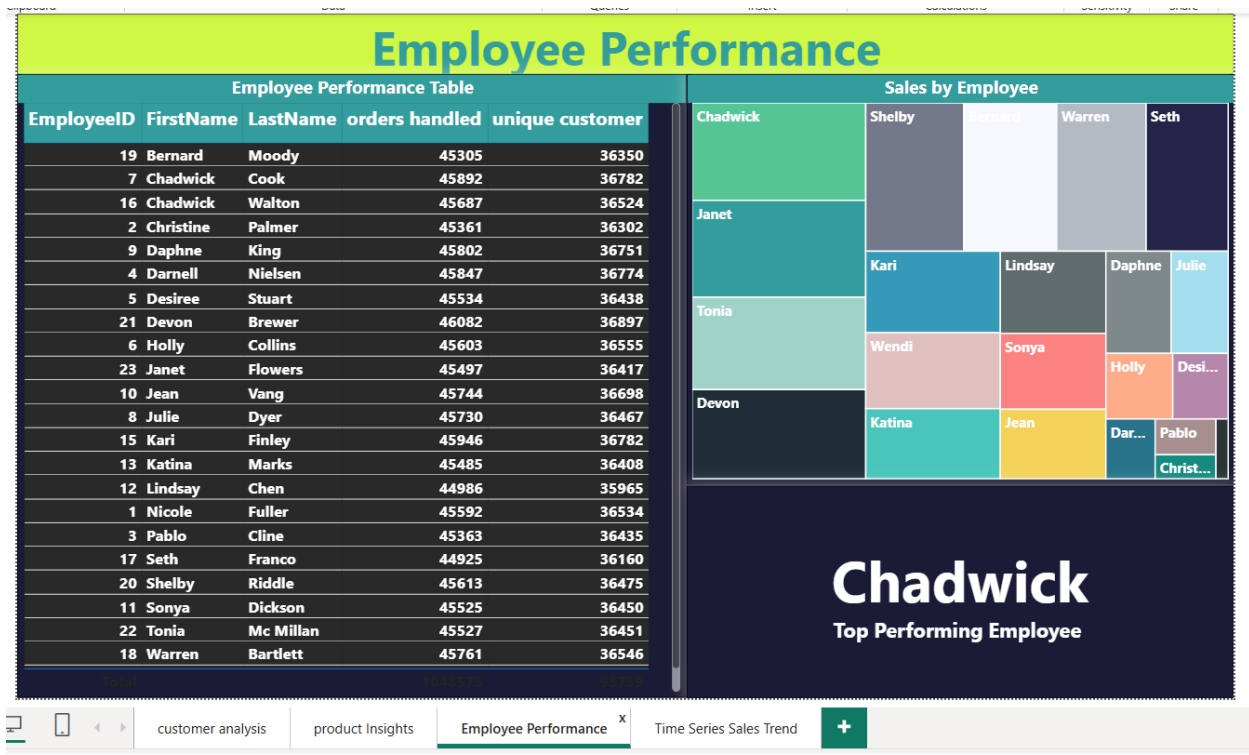
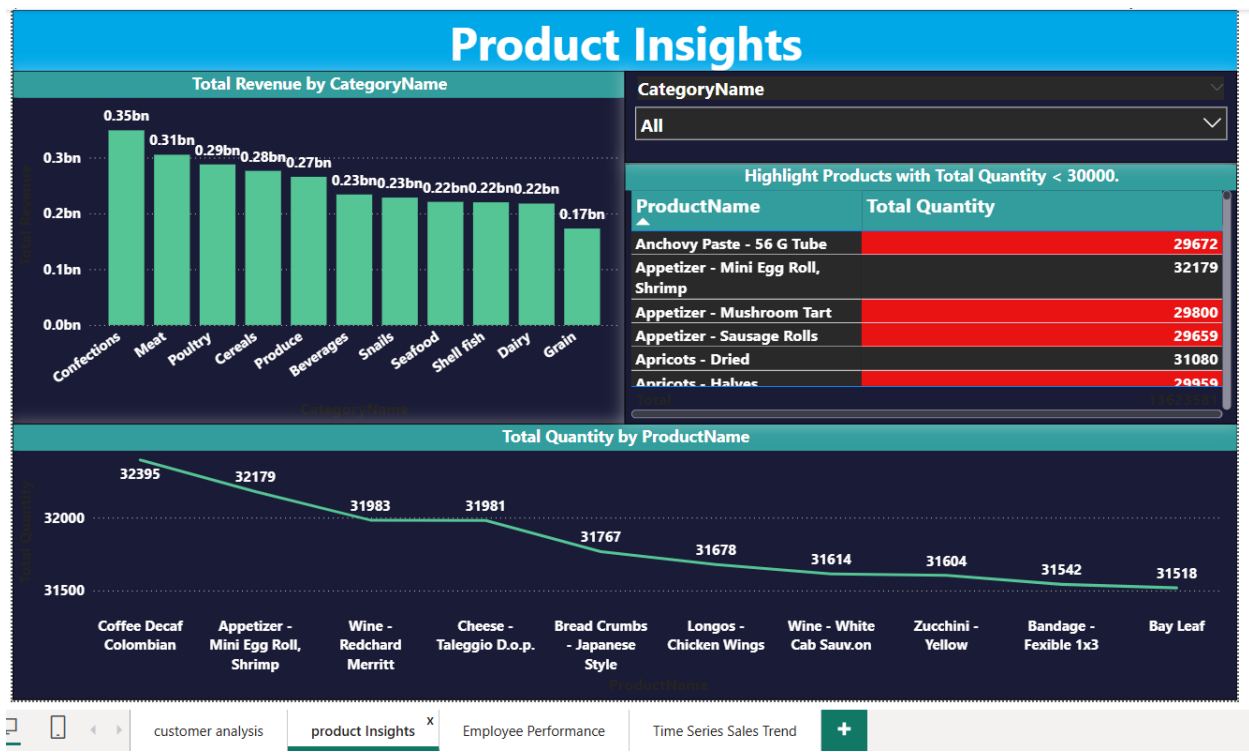
Use slicers for:

- Country
- Category
- Customer Segment

Use **Smart Narrative** and text boxes to describe insights.

Output :-





Time Series Sales Trend

Year-to-Date Revenue

3bn

Total Revenue by Month



customer analysis

product insights

Employee Performance

Time Series Sales Trend ^x

